

The General Architecture of Languages Holds Key to Consciousness

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Abstract. If we are to truly understand consciousness, we have to account for the subjectivity of Self, the sense that we all have of being separate from others, from our own bodies and from the physical environment that surrounds us. How is it that I 'know' that I am not you, my body, the doorjamb, or the trees outside my window?

I argue in this paper that this subjective sense of separateness comes to us courtesy of an unheralded design feature common to all languages. That design feature is the bifurcation of concepts into meanings that represent end points on a spectrum of meaning. These meanings are spatially separated but remain connected in the network of meanings that make up any Meaning System (or language). Such a bifurcation of meaning along a spectrum is a condition precedent to consciousness because there can be no awareness or understanding of one pole without the other. 'Yes' and 'no', for example, only have meaning in relation to each other. They are not absolutes. The same is true of 'night' and 'day', 'up' and 'down', 'good' and 'evil', 'life' and 'death', and all other meaning polarities. This architecture necessarily underpins all languages for without it there would be no meaning.

The subjective 'I' sits at the center of this world of meaning polarities. An outgrowth of the Me—Not Me continuum, it like all other polarities is a construct of languages. Since the physical senses are mere processors of data, it falls on the inner 'I' to synthesize and interpret incoming sensory information. It is this metaphysical construct that actually sees, hears, feels, tastes and smells. In this sense all languages are organs of perception because they organize the world of sense perceptions around what is 'Me' and what is 'Other'. Without this polarity there would be no consciousness. In the words of Carl Jung, the renowned Swiss psychologist and psychoanalyst, "[C]onsciousness presupposes a differentiation into subject and object and a relation between them. Where there is no 'other', or it does not yet exist, all possibility of consciousness ceases."(1)

Languages, by way of this forgotten, self-similar architectural design, have fabricated the sense of 'I'-ness out of nothing. Like pulling on a VR headset, the acquisition of a language transports us into an entirely different cognitive realm, a domain in which we feel separate from others and from our surroundings. The litmus test for consciousness is not whether an entity we see from a distance is physically distinguishable from its surroundings, but rather that 'It' perceives itself as such. A pre-linguistic infant, not yet separated emotionally from its mother, is not conscious. The child will not make that transition until it builds an internal sense of self by activating and exercising the 'No' reflex.

This presentation will offer compelling evidence that this unique design feature of languages is the constituent ground of, and explanation for, the phenomenon of consciousness. (1) *Aion: Researches Into the Phenomenology of the Self*, p.193. (Princeton University Press, 2nd Ed.)

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Introduction

The question of why we are conscious has been described as the hardest problem in science and philosophy. Why is it that human behavior is accompanied by subjective experience? In a TEDTalk given in 2014, David Chalmers suggested that radical ideas might be needed in order to integrate consciousness into science. He talked about two such radical ideas in particular: First, that consciousness might be a fundamental building block of nature alongside space, time, mass and charge. Second, the idea that every system is conscious to some degree. This second idea is known as Panpsychism.

My radical idea is that consciousness is a construct of human languages. And I should emphasize here that I am talking about all human languages. All languages contain the same design feature that makes consciousness an inevitable outcome of learning them. If this is true, and I believe it is, then the consciousness you take for granted would collapse without languages. For the purposes of this talk I will define consciousness as ‘knowing that you know something’. This is the logic of consciousness. So, for example, I know that I am giving a talk on the role languages play in making you conscious, and I know that I know I am giving this talk.

In this presentation, I propose to look at the architectural design in languages that gives rise to consciousness from two perspectives: In Part I, I argue that it is the acquisition of a language that moves us from an unconscious state to a conscious one. In Part II, I suggest that we need to radically alter our perception of what a language is in order to understand the role languages play in creating consciousness. It requires a paradigm shift. Also in Part II, I’ll address the specific design feature of languages that gives rise to consciousness. I will conclude the presentation with a thought experiment that demonstrates how removing this internal design feature of languages from our suite of cognitive abilities collapses consciousness.

Part I: Languages and Consciousness

In my abstract for this talk I posed the following question: ‘How do I know that I am not you, my body, the doorjamb or the trees outside my window?’ Or, to bring the question into the present and make it specific to each one of you here, ‘How do you know that you are not the person sitting next to you?’ On its face, this sounds like an absurd question. Of course you know, although almost certainly you don’t know how or why you know.

You see yourself as separate from each one of the aforementioned entities – your body, a doorjamb, trees and the person sitting next to you (each a ‘**Spacial Entity**’). Spacial entities are just entities that take up two or three-dimensional physical space. But you also see yourself as separate from this or that action, event, or time period (each a ‘**Temporal Entity**’). Temporal Entities are

actions, events or time periods that take place over time. They have a beginning and an end. This would include, for example, me raising my arm, or a sound that persists for a certain span of time. Other examples might be a period of time such as a day, or a year, etc. The point to be made is that you not only perceive yourself as being separate from all Spacial Entities, you also perceive yourself as separate from all of Temporal Entities.

The reason I bring up this perceptual divide between things 'Me' and things 'Other' is that it cuts to the heart of who we are as conscious beings. Each of us knows that we stand apart from all of the things we see out there in the world, and all actions or periods of time that have a temporal existence. And this is the essence of consciousness. We are very clear about the boundaries between 'What is Me' and 'What is Not Me'. I will refer in this presentation to things that are 'Not Me' as 'Others.' This includes all Spacial and Temporal Entities that are identified as Not Me.

But all of this raises an interesting dichotomy. If you look at the timeline of your life, and if the litmus test for consciousness is knowing that you are not any of these things out there that I am calling Others, then there clearly was a time you did not possess that degree of self awareness; a time when could not distinguish between what was you and what was not you. That certainly was the case at conception but also at the time of your birth. So either the lines were blurred or they did not exist at those junctures. And let me hasten to add here there was nothing wrong with the function of your eyes or ears, or for that matter any of your other senses at those times. So what happened between then and now to cause such a radical difference in our perception of boundaries and reality?

My contention is that language happened. Somewhere between birth and the present time, we acquire our first language – our mother tongue – and a system of meaning (a '**Meaning System**') is installed, or more precisely is activated. In Part II of the presentation, I will talk about how this understanding requires a complete reframing of how we see languages. The installation of a Meaning System causes a separation to occur between each of us and the rest of the world. Initially a wedge, it later becomes a gulf. And the effect of acquiring a Meaning System is to tease apart the Subject 'Me' (and later 'I') from all Others, to the point that we each perceive things today – a world of conceptual entities distinct from ourselves.

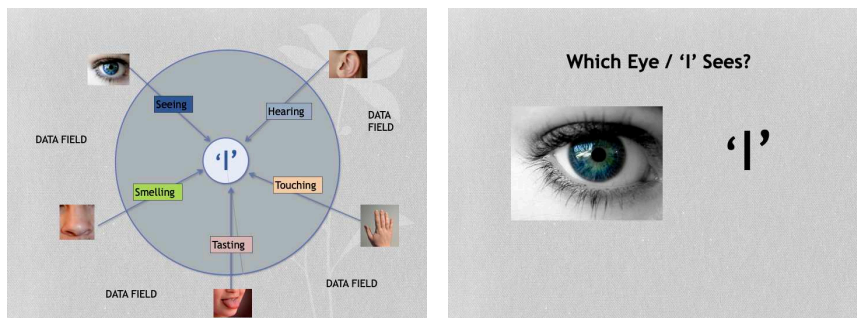
My claim is that it was, and is, this Meaning System — installed in the mind/brain when you were acquiring a language — that caused you to move or transition from an undifferentiated being to a differentiated, conscious one. Later, I will explore the mechanism that allows this transition to happen. Of course this transition was completely seamless and unconscious and remains out of your conscious mind today. Hence our perception that consciousness is the most mysterious phenomenon in the universe, the so-called 'Hard Problem' of consciousness, and so on.

The following quotation, taken from Carl Jung's book *Aion*, provides a psychological understanding of how the phenomenon works. According to Jung:

Union of opposites is equivalent to unconsciousness, so far as human logic goes, for consciousness presupposes a differentiation into subject and object and a relationship between them. Where there is no 'other,' or it does not yet exist, all possibility of consciousness ceases.¹

Effectively then, in the terms that I have been discussing the movement from unconscious to conscious, prior to the acquisition of a language there was a unity of opposites – *i.e.* there was no opposition (or distinction) between what would become you and what you would later identify as 'Other.' You were identified with or identical to these Others. And this identity of opposites, as Jung calls it, precludes consciousness. Incidentally, this is the explanation of why we have no memories prior to a certain point in our personal narrative. The reason is that memories cannot attach to an undifferentiated being. After the installation of a Meaning System and the separation of subject and object, you became differentiated and conscious.

You might have been moved in response to my earlier question – as to how you know the person sitting next to you is not you – to assert that you can 'see' with your own eyes that you are not the person sitting next to you. And you would be right, but for the wrong reason. It is not the eyes that allow you to see. The eyes see nothing. They are merely receptors that process light. The same is true of the other senses. Seeing is actually a percept of the mind/brain. You 'see' with the mind/brain – with the inner 'I' shown here in the accompanying slide.



Hearing, touching, tasting and smelling are all percepts of the mind. It is the inner 'I' that organizes and interprets the information or data that is being piped in from the senses.

¹ *Aion: Researches into the Phenomenology of the Self*, C. G. Jung (Princeton University Press, Second Edition), translated by R. F. C. Hull, at p. 193.

Prior to acquiring a language you could not see these ‘Others’ because you did not have the capacity to see with this inner ‘I’. Why? Because it had not yet been installed. What allows you to see is the installation of a Meaning System that serves to separate ‘What is Me’ from ‘What is Not Me’. Languages are therefore very much, from this perspective, **Organs of Perception**. I don’t mean to imply that the movement from the undifferentiated to differentiated – *i.e.* unconscious to conscious – happens all at once. Rather, it is a process that is reinforced over time. Nonetheless, the process happens rather quickly as educators and parents will readily attest.

From a behavioral standpoint this transition takes place in the so-called terrible or terrific twos, depending upon how you view this developmental stage. This is the time when you are finding your feet as a separate being and exercising your inalienable right to say ‘No’. The psychological impact of exercising the ‘No’ reflex is to separate you from those around you. In effect, you are saying ‘Don’t mistake me for you!’ ‘I am not you!’ And in the exercising the ability to say No, you are creating a Self behind the ‘No’. You are in the beginning stages of recognizing that you are a One (a distinct entity) in a world of Ones.

Part II: Languages as Meaning Systems

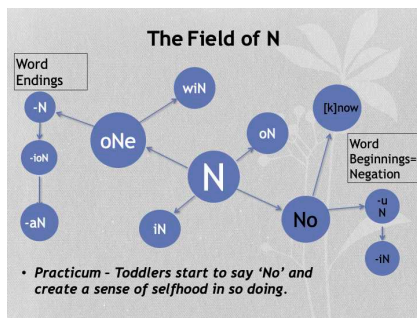
In Part I of the presentation, I tried to show that there was a time that you were not conscious and that it was the acquisition of a language – your mother tongue – that moved you from an unconscious being to a conscious one. Now I want to address how that happens, first in the big picture and then more specifically addressing the mechanism itself.

(A) A Paradigm Shift: My talk is premised on a radical reframing of how we view the nature of languages. You probably have some idea of a language as a tool used by human beings to communicate. Perhaps you see languages as a corpus of words held together by rules of grammar and syntax. I want to suggest that languages are maps to meaning (**‘Meaning Maps’ or ‘Maps’**) that reside in the unconscious. We know that the process of making meaning happens in the unconscious. We know too that the process is not amenable to introspection.

My argument is that while you are busily learning a language as a toddler, you are also laying down a Meaning Map in the unconscious, a process that we are entirely unaware of. However, the fact that this hidden Map is unconscious does not mean that it does not exist. A dramatic illustration of the robust quality of the Map is that I can make up a completely outrageous sentence that has never been uttered in the history of the world and, if we speak the same language, you will have no problem understanding me. In this vein, here is my example:

The pink giraffe, who wore a knitted green sweater with purple tassels, roller bladed down the Los Angeles freeway, weaving in and out of the rush-hour traffic, all the while singing, ‘I Did It My Way’ at the top of her lungs.

Has this sentence ever been uttered in the history of the world or the universe?
Personally, I doubt it!



Another way to think about this Map is that it is an integrated network of meaning, something like the World Wide Web in terms of connectedness and connections. Just to give you an idea of what the Meaning Map in English might look like – at least a fragment of it – here’s a diagrammatic representation taken from words whose meanings are built with the consonants ‘N’ and ‘T’. So for example, words meaning ‘connection’

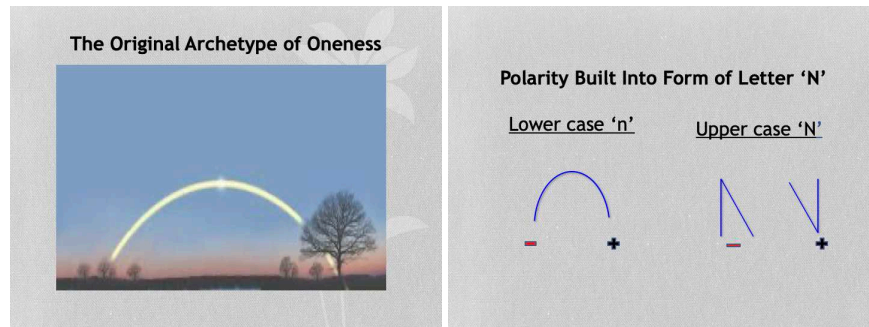
include [k]NiT, [k]NoT, uNiT and uNiTe. Words meaning ‘separation’ include uNTie, aNTi and NoT. And words which carry the meaning ‘between’ include iNTer, iNTra, eNTer and NeuTral.

What are the properties of this Map to Meaning? The key thing to notice is that **Meaning is arranged in polarities**. This is key because it is this design feature that gives rise to both meaning and to consciousness. Further, the Map is **internal** and **self referential** in the sense that words point to concepts in the Meaning System, not to things in the world. Concepts therefore precede any particular examples in the World. A squirrel is so-called because in the unconscious Meaning Map the word ‘squirrel’ carries the meaning ‘twitchy’ and ‘restless’.

How does this interpretation differ from the conventional understanding? Well, in short, we have taken what were once descriptions of things and turned them into names that reference things. But names are simply descriptions of the entity named based on a distinguishing trait. Here are some other examples: A sheep is a follower; a window is a wind-hole (similar to wind-screen); a woman is a womb-man or a man with a womb; and a tree gets its name because trees embody the principle of ‘three’ in their design (each node is the intersection of three branches). In the same way, an ‘emergency’ is a description of what an emergency is, namely an ‘emerging urgent agency’. And a neighborhood is just a near (neigh or nigh) living (bor) district (hood).

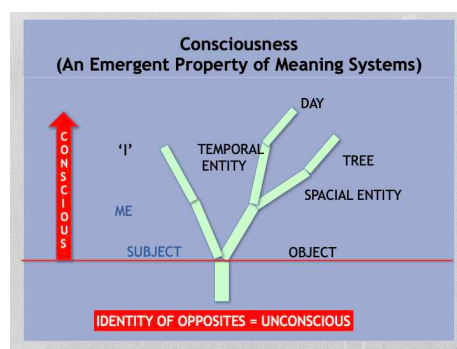
What is a name? To reiterate, a name is a description of something based upon a distinguishing property or quality of the entity being described. The name ‘One’ stands for the concept of polarity inherent in all individuated entities (Ones). On this basis, can you guess what the symbol for the concept ‘One’ might be? I have shown it here in the accompanying slide. It is the arc of the sun as it passes through the sky from dawn to dusk. This is the original arc(h) type (or archetype). What symbol could more perfectly describe the concept

‘One’ than the path of the sun through the sky? The defining quality of Oneness in all its manifest forms is that it embodies polarity. Without polarity, the concept of One-ness is impossible. Anything that has this defining quality – either spacial or temporal – necessarily exhibits polarity.



That is why we use the symbol n (or N) to represent the concept of One-ness. In both its lower case and upper case forms it embodies the principle of polarity, as shown in the figure to the right above. So it turns out that the defining quality of oneness itself is coded into the alphabetic letter symbol that is used by all languages that utilize the Roman alphabet as the basis of their writing system. What does this mean in regard to how we see languages? In short, it means that meaning inheres in the system of sounds used by a language system. Meaning is generated by each such language system. All languages are therefore webs of Meaning. The implications of this, if correct, are massive for consciousness, for the origins of languages and for how information is stored in the brain, all of which are regarded as mysteries at the present time.

The accompanying slide shows a schematic representation or skeleton of one such Meaning Map for an unspecified language. It could be any one of the 6,500 or so Meaning Systems represented on Earth. It is another way to look at Jung's preconditions – subject, object and relation between the two. In the lower portion of the tree you can see the how the lack of opposition between subject and object precludes consciousness. (Unity of Opposites). Above the red line, with the separation of subject and object, is the zone of consciousness. The bifurcation point represents the genesis of consciousness. These branches on the tree then take on the quality of Me vs. Other and eventually 'I' vs. All Named Others.



You might recognize this tree from the Bible, specifically Genesis, as the Tree of the Knowledge of Good and Evil. In Genesis, God warns the newly-formed man not to eat from the Tree in the Middle of the Garden of Eden or he will die. But the serpent explains to his partner that eating from the Tree will make humans like God(s), knowing the difference between Good and Evil. Both of them then eat from the Tree. This is the genesis of consciousness. The fruit of the Tree *is* Consciousness. Before this rupture, the man and the woman felt no shame. Afterwards, they realize they are naked. In other words, they become self aware. Note too that before the Fall they are simply called ‘man’ and ‘woman’. After the Fall, they are called Adam and Eve.

I would argue that the narrative in Genesis uses the polarities of Good and Evil as exemplars of a much more general principle, namely, that it is **Polarity of Meaning** which leads to consciousness. This, I think, is the main takeaway of the narrative. And this polarity is necessarily built into the structure of all languages – necessarily so, because without that architecture you would not be conscious. Let me turn my attention now to how the architecture of the Meaning System is defined by polarity.

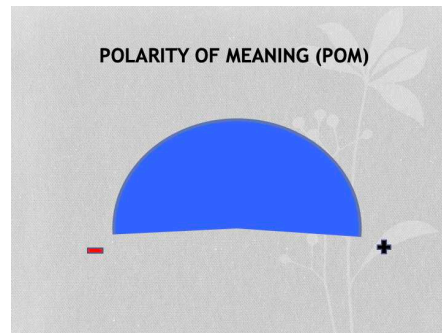
(B) Polarity and Meaning: Meaning, as it pertains to a Meaning System, is a term of art. It does not have the usual meaning of the word ‘mean’ in the English language: What do you mean?; you are mean!; that gesture means so much to me; etc.). In a Meaning System, meaning only exists in relation to its opposite or complement. Another way of saying the same thing is that absent polarity there is no Meaning.

Why is this? Because concepts don’t exist in some absolute sense. They only exist in relation to one another. You need one pole of the axis to ground the other. There is no up without down, left without right, right without wrong, life without death, and so on. Whenever you engage one pole of the axis of the Meaning System, the other pole comes into play. V.S. Ramachandran, the eminent neuroscientist, makes the same point with regard to the Self and Qualia in the following quotation:

And [Francis] Crick and [Christof] Koch have argued, first let’s solve the qualia problem and let’s get to the self problem later, and I’m saying that’s impossible. With all due respect to Francis and Koch . . . I’m saying there is no earlier stage called qualia and subsequent stage, self inspecting the qualia — there’s no such thing. And the reason is very simple, there is no such thing as free-floating qualia. It’s an oxymoron without a self experiencing it. Likewise, a self without qualia, without any sensations, memories, subjective sensations, is meaningless. So I claim that these two co-evolved in evolution . . .²

² The Science Studio: Take the Neuron Express for a brief tour of consciousness. Interview with V.S. Ramachandran, dated June 10, 2006. (The Science Network).

The intellect buys into a belief, which it holds dear, that these poles of Meaning can be isolated from each other. But this is an illusion. When you think of the word 'Man' you know what it signifies only in relation to its opposite. Somewhere in the unconscious when you say or think the word Man it is being compared to the concept Woman (or perhaps Animal, depending on the context). When you think of going up, it is being unconsciously grounded by the idea of going down, and so on. Thus polarity is built into (embedded in) the architecture of every Meaning System. And every concept in a Meaning System can be represented by a plus and a minus (to indicate polarity).



We can represent this relationship as shown in the accompanying diagram. This is the coding for the archetypal One – which all entities or Ones must be in order to be seen and named. What is the takeaway from this? Polarity is not a property of the *world out there*. Polarity is the prism or filter through which we view the world. The ultimate polarity is the axis that separates from Me (and the subject 'I') from all Others out there. I say ultimate because it is this separation from Others that gives rise to consciousness.

The sum total of all these axes defines a Meaning System, and the entire fabric of all these polarities is woven together into a way of 'seeing' the world. The depiction of Neith (or NiT), an ancient creator goddess in Egypt, demonstrates that the ancient Egyptians were familiar with this fact. Her name is made up of two consonants, N and T. The wave sign (N in her name) represents the undifferentiated and unseen world (the unmanifest or negative vector); the sign underneath it (T in her name) represents the manifest physical and temporal world (the + vector). The goddess creates the world by weaving these two worlds together with her weaving shuttle (depicted to the right of the other two symbols).



The important thing to recognize is that You, Me – All of Us – are immersed in a world of Meaning. Every day, all day, we are surrounded by these Polarities of Meaning. We live in a world that comprises up-down, right-left, big-small,

stationary-moving, light-dark, wrong-right, erudite-ignorant, hot-cold, and every other polarity. And yet we don't give this a moment's thought. It is the reality we know and have always known, as long as we have been conscious beings. And amid all of this roiling, teeming fabric of Meaning, we stand irreducibly, at the center of our own experience and our own reality – a ONE in a sea of ONES (an entity among other entities). It is as though, with the acquisition of a language, we pulled on a virtual reality headset that modified our reality. The acquisition of a Meaning System casts us into another cognitive space altogether; a space and reality that we have been living in ever since. This of course goes a long way to explaining why we are unaware of this fabric of Meaning. We simply have no other reference point. The reason is that prior to being colonized by a Meaning System, neither you nor I were conscious. We are like fish that swim in a medium of which we know nothing.

Conclusion

In conclusion, I would like to leave you with this thought. Given this understanding of how a Meaning System is constructed, imagine what would happen if you could remove this headgear (VR headset) at night? Let me illustrate with a thought experiment: Imagine a hand-held fan like the one shown in the illustration below. The fan is a metaphor for the arc that mediates all polarities. Now imagine that each of the polarities present within a Meaning System; Pain—No Pain; Before—After; Today—Tomorrow; Wide—Narrow; Near—Far; Sick—Healthy; Alive—Dead; and so on, is represented by a fan like the one shown here. If you now proceed to close each one of these fans – the equivalent of closing down every one of these polarities – you eliminate the polarity that existed when the fan was open. In the same way, shutting down the polarities that inhere within a Meaning System ultimately has the effect of collapsing consciousness.



Why is that? The reason is that our 3-D vision, internal narrative, emotions, and all the rest of our subjective experiences that make us conscious are mediated by these polarities. They are mediated by the third eye (or inner 'I') that is a product of a Meaning System. By removing these polarities, you therefore gut reality on two levels:

(1) You collapse the distinctions between 'What is Me' and 'What is Other'; and

(2) You collapse the distinctions between one 'Other' and another 'Other' in the system of 'Others'.

Although these distinctions appear to be rooted in our sense perceptions, that is emphatically not the case. The differentiations that we make are the product of a Meaning System rooted in polarity, naming and the descriptive differences between 'Others'. The following truths are apparent:

**(1) THERE IS NO MEANING OR CONSCIOUSNESS
WITHOUT POLARITY**

**(2) MEANING IS CONSCIOUSNESS /
CONSCIOUSNESS IS MEANING**

(3) CONSCIOUSNESS $\equiv$ MEANING